

Technical Operations & User Manual - Machine M102 (5-Axis CNC Milling Center)

1. Machine Specifications & Environmental Limits

Machine Model: M102 High-Precision 5-Axis CNC Milling Center.

Spindle Power: 35 kW liquid-cooled high-speed spindle (up to 24,000 RPM).

Location: Bay 3 - Precision Machining Wing.

Operating Temperature Threshold: Normal operating temperature is between 20 deg C and 45 deg C. The thermal limit for the primary spindle and axis drives is 65 deg C.

2. Overheating Causes and Thermal Shutdown

Overheating of Machine M102 is primarily caused by coolant circulation failure, blocked radiator heat-exchanger fins, coolant pump pressure drops below 2.5 bar, or prolonged heavy-load roughing passes without sufficient dwell time.

When spindle temperature exceeds 65 deg C, the internal thermal protection sensor triggers an automatic Emergency Thermal Shutdown (Fault Code E-704).

In the event of an overheating shutdown, operators must NOT force a reboot immediately. The machine must be allowed to cool down to 35 deg C, and the radiator coolant loop must be flushed and inspected for sediment or particulate blockages before re-engaging the spindle.

3. Cooling System Architecture

The closed-loop cooling system uses a 50/50 mixture of demineralized water and ethylene glycol industrial coolant.

Coolant tank capacity is 60 Liters with an integrated inline filter rated at 25 microns.